Damping from the Escape Probability: a free calibration rule for diffusion-based reachability ranking in uncertain graphs

Ioannis Konstantellos
Computer Engineering and Informatics Department
University of Patras, Greece

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Abstract

Damped diffusions—personalised PageRank and its variants—are a standard cheap heuristic for ranking vertices of a probabilistic graph by expected reachable mass. Their damping factor α is almost always inherited from web search (0.85) or tuned by grid search against ground truth that is itself #P-hard to compute. We show that neither is necessary.

Our contribution is a closed-form calibration. The best-performing α is predicted by the *mean escape probability* $\bar{P}_{\mathrm{esc}}$, the average probability that a vertex has at least one incident edge present, which is already computed when the transition operator is built and is therefore free. Fitting $\alpha = \mathrm{clip}(0.502 + 0.558\,\bar{P}_{\mathrm{esc}}, 0.4, 1)$ on 54 training instances gives Pearson $r = +0.965$ and out-of-sample mean Spearman $\rho = 0.894$, recovering 98.2% of a per-instance oracle. The rule beats *every* fixed α, including the best one selectable with hindsight on the test set (0.894 against 0.677), because instance densities vary enough that no single constant serves them. It is a property of the problem rather than of one operator: refit independently for standard push personalised PageRank it gives $\mathrm{clip}(0.241 + 0.859\,\bar{P}_{\mathrm{esc}})$ with $r = +0.961$ and 98.7% of that family's oracle. It survives a 14× change of scale without refitting, and on real citation networks it still beats the best constant α, at 91% of oracle.

We also situate the escape probability itself. It is not a new score: it is exactly the correlation-free ($\eta = 1$) zeroth iterate of Karrer, Newman and Zdeborová's message-passing percolation, and its value is purely one of engineering. Running that recursion properly is better—on 30 real dense instances converged message passing reaches $\rho = 0.9931$ against 0.9852 for the free iterate—and both dominate a 200-sample Monte Carlo (0.9741) at a seventh of its cost. The ranking signal saturates after two iterations. A reverse-reachable-set estimator from the modern RIS lineage, evaluated in its singleton-spread form, does not change this picture: at matched budget it trails plain possible-world sampling for whole-graph ranking, an information-per-sample effect we quantify and locate. Finally we report a negative result: routing among heuristics by $\bar{P}_{\mathrm{esc}}$ works on synthetic graphs and fails on real ones, which we take as evidence that cheap global density features do not support algorithm selection here. Code and data are provided.

1 Introduction

Let $G = (V, E)$ be undirected with $|V| = n$, each edge (i, j) present independently with probability $p_{ij} \in (0, 1]$, and let $x \in \mathbb{R}^n_{\geq 0}$, $\sum_i x_i = 1$, be a prior over which vertex is the target of a routing or search operation. Sampling a world G' and writing $C(i)$ for the connected component of i, we rank vertices by

$$y^*_i \;=\; \mathbb{E}\Big[\textstyle\sum_{v \in C(i)} x_v\Big], \tag{1}$$

the expected prior mass reachable from i.

What this quantity is. Equation (1) is a known object under two other names, both verified numerically in Section 4.

Proposition 1 (Influence spread)**.** *Under the live-edge equivalence of Kempe, Kleinberg and Tardos [34], a vertex is activated by the Independent Cascade model from seed set S exactly when it is reachable from S in a sampled live-edge graph. For $S = \{i\}$ on an undirected support reachability is connectivity, so y_i^* is the node-weighted single-seed expected spread with weights x.*

Proposition 2 (Percolated component centrality)**.** *With uniform prior and constant edge probability, $n \cdot y_i^*$ is the Edge Percolated Component score of [44, 20].*

Evaluating (1) exactly subsumes two-terminal network reliability and is #P-hard [61, 54], so the field relies on fast alternatives: structural proxies [17], path truncation [16], linear-system heuristics [31], sampling estimators with guarantees [10, 59, 60, 21], and—the family this paper is about—damped diffusions.

The gap. A damped diffusion interpolates between the prior and a propagated signal,

$$y \;\leftarrow\; (1-\alpha)\,x \;+\; \alpha\,\mathcal{T}y,$$

for a propagation operator $\mathcal{T}$. Personalised PageRank [50, 27] is the canonical instance and variants are routinely applied to uncertain graphs [35, 51]. The damping factor controls how far signal travels before being pulled back toward the prior, and it is consequential: across our held-out instances, moving α over its range changes mean Spearman ρ by more than 0.2 for both operators we study.

Yet α is usually not chosen. It is inherited from web search, fixed by convention, or—when performance matters—grid searched against ground truth that is exactly the expensive quantity the heuristic exists to avoid. The same convention governs current practice: a recent survey of efficient personalised PageRank computation treats α as a given constant throughout [65], and PPR-based graph neural networks either fix the teleport weight or learn propagation weights from labels [18, 8, 52]. The damping literature analyses sensitivity to α [9, 11] and ties it to a target conductance in local partitioning [2]. The closest precedents derive an optimal α in closed form from the parameters of a *planted model*: Kloumann, Ugander and Kleinberg [36] obtain $\alpha^* = (p_{\text{in}} - p_{\text{out}})/(p_{\text{in}} + p_{\text{out}})$ for seed expansion in a stochastic block model, Avrachenkov, Kadavankandy and Litvak [5] an optimum from the same model's density ratio, and Li, Chien and Milenkovic [41] optimal generalized-PageRank weights under related generative assumptions. While prior work has theoretically attempted to estimate planted model parameters from the graph (e.g., using the Allman et al. estimator [1]), such methods require known block sizes and are notoriously brittle on non-SBM graphs. In contrast, our $\bar{P}_{\text{esc}}$ calibration operates directly on the topology of any real network without requiring block assumptions, and is computed for free when the operator is built.

Contributions.

1. **A free damping calibration** (Sections 3–5), the main result: α predicted from $\bar{P}_{\text{esc}}$ at $r = +0.965$, recovering over 98% of a per-instance oracle, beating every constant including the hindsight-best, transferring across operator families and across a 14× change of scale, and robust to rounding its coefficients.
2. **An honest placement of the escape probability** (Section 6). P_{esc} is the $\eta = 1$ zeroth iterate of message-passing percolation [32]; we say so, implement the full recursion as the proper deterministic competitor, and show the ranking signal saturates after two iterations while both dominate small-budget sampling. We also benchmark a reverse-reachable-set estimator from the modern RIS lineage [10, 60, 24, 25] in its singleton-spread form (Section 7), and locate why it trails the possible-world estimator for whole-graph ranking at matched budget.

3. **A negative result on algorithm selection** (Section 8). Routing among heuristics by $\bar{P}_{\text{esc}}$ beats the best single method on synthetic graphs and fails outright on real citation networks. We report the failure rather than the synthetic-only success.

2 The escape probability, and what it is

Definition. *$P_{\text{esc}}(i) = 1 - \prod_{j \in N(i)}(1 - p_{ij})$ is the probability that vertex i has at least one incident edge present—that it is not isolated in a sampled world. We write $\bar{P}_{\text{esc}} = \frac{1}{n}\sum_i P_{\text{esc}}(i)$.*

The expression is not new and we make no claim on it. It is the one-hop case of the activation recursion in MIA/PMIA [16], the complement of the node isolation probability studied in wireless connectivity [58], and the per-node summand of the giant-component size in percolation theory. Most precisely:

P_{esc} is the correlation-free limit of message-passing percolation. Karrer, Newman and Zdeborová [32] compute, deterministically, the probability that each vertex belongs to the giant cluster. In membership variables their recursion reads

$$\nu_{i\to j} \;=\; 1 - \prod_{k \in N(i)\setminus j} \big(1 - p_{ik}\,\nu_{k\to i}\big), \qquad \mu_i \;=\; 1 - \prod_{k \in N(i)} \big(1 - p_{ik}\,\nu_{k\to i}\big). \tag{2}$$

Initialising $\nu \equiv 1$ and evaluating μ once, with no message updates, gives $\mu_i = 1 - \prod_k (1 - p_{ik}) = P_{\text{esc}}(i)$ exactly. The escape probability is therefore the zeroth iterate of (2) under the assumption that every present edge leads to the giant cluster. Equivalently, for constant p and degree k it is the $u \to 0$ case of the $1 - u^k$ membership probability in the generating-function formalism of Newman, Strogatz and Watts [47] and Callaway et al. [12]. We verify the identity numerically (Section 6).

The consequence for this paper is a demotion and a clarification. P_{esc} is not a mechanism we discovered; it is a known quantity in its most degenerate form. Its interest here is purely one of engineering: it costs one pass over the edge list, which building any transition operator already performs, so it is free in a sense that no iterated quantity is. That is why we use it as the feature for the calibration rule, and it is the only property of it we rely on. $\bar{P}_{\text{esc}}$ is also a smooth, bounded measure of *effective* density—it approaches 0 when vertices are typically isolated and 1 when they are typically connected, and unlike average degree it accounts for edge reliability.

3 The calibration rule

Let W be row-stochastic on the potential adjacency with $W_{ij} \propto -\ln(1 - p_{ij})$ (Section 5 shows the weighting is immaterial). We study two damped diffusions: standard push personalised PageRank [50, 27],

$$y \;\leftarrow\; \text{normalise}\big[(1-\alpha)x + \alpha\, W^\top y\big], \tag{3}$$

and an escape-scaled pull form in which each vertex is attenuated by its own escape probability, making the operator sub-stochastic,

$$y \;\leftarrow\; \text{normalise}\big[(1-\alpha)x + \alpha\,(P_{\text{esc}} \odot (Wy))\big], \tag{4}$$

a mild instance of the partially absorbing random walk [63] with absorption set by edge reliability. Both need an α, and the value that maximises rank correlation varies systematically with how connected the sampled worlds tend to be. Sparse instances need small α: components are small and y^* is close to x. Dense instances need large α: the prior is nearly irrelevant and the propagated structure carries the signal. $\bar{P}_{\text{esc}}$ measures exactly this axis.

We calibrate on 54 instances at $n = 64$ (Erdős–Rényi, Barabási–Albert and grid topologies; six probability ranges; three seeds). For each we grid search the α maximising Spearman ρ against Monte-Carlo ground truth, then regress it on $\bar{P}_{\text{esc}}$. Least squares gives

$$\boxed{\alpha \;=\; \text{clip}\big(0.502 + 0.558\,\bar{P}_{\text{esc}},\; 0.4,\; 1.0\big)} \tag{5}$$

with Pearson $r = +0.965$ ($p = 5.9 \times 10^{-32}$), Spearman +0.959, and residual RMSE 0.043 (Figure 1(a)). The lower clip never binds; it is retained for degenerate inputs. Since $\bar{P}_{\text{esc}}$ is available at initialisation, (5) adds no measurable cost.

4 Experimental setup

Ground truth. Monte Carlo over sampled worlds, extracting connected components with SciPy's sparse graph routines; 3,000 samples per held-out instance, 2,500 during calibration. Significance is paired Wilcoxon signed-rank; confidence intervals are bootstrap percentile intervals over 20,000 resamples. Timings are single-core wall clock.

Identity checks. On an $n = 60$ instance the maximum discrepancy between y^* and an independently simulated single-seed Independent Cascade with node weights is 0.0058, against Monte-Carlo noise of $1/\sqrt{4000} = 0.0158$. With uniform prior and constant p, $n \cdot y^*$ reproduces the EPC score at Spearman 1.000000. Propositions 1 and 2 hold as stated.

Synthetic instances. A held-out set of 100 instances at $n = 100$ (four topologies including Watts–Strogatz, absent from calibration; five probability ranges; five seeds). A stratified panel of 400 instances at $n = 900$ built for coverage: $(0, 1]$ divided into eight equal-width strata of $\bar{P}_{\text{esc}}$ with 50 instances each, and within a stratum every other factor randomised (topology, average degree over $\{4, \ldots, 12\}$, probability range, prior type, seed). Design matters here and we can show it rather than assert it: operators (3) and (4) are indistinguishable under the $n = 100$ grid design ($p = 0.60$) and separated at $p = 1.5 \times 10^{-21}$ under the stratified one; running the identical stratified design at $n = 100$ reproduces the separation ($p = 1.7 \times 10^{-10}$), so the cause is balanced coverage of the density axis, not scale.

Real networks. Three genres, seven base graphs. Citation: Cora ($n = 2708$), CiteSeer (3279), PubMed (19717) from the Planetoid release plus their k-cores. Co-purchase: Amazon Computers (13752) and Photo (7650). Co-authorship: Microsoft Academic CS (18333) and Physics (34493). Degree heterogeneity is far beyond the synthetic families (coefficient of variation 0.77–1.97) and clustering runs 0.14–0.40 against ≈ 0 for Erdős–Rényi. Edge probabilities are assigned—no dataset carries observed p_{ij}—by uniform ranges and by the weighted-cascade rule $p_{ij} = 1/\max(\deg_i, \deg_j)$, which is out of distribution for our rule. Subgraphs are snowball samples, which preserve local density where a random node sample would artificially sparsify.

5 Results: the calibration

5.1 No fixed damping factor is adequate

Figure 1(b) sweeps a single α applied uniformly to all 100 held-out instances. The best attainable value is $\rho = 0.677$ at $\alpha = 0.925$ for the escape-scaled operator and 0.736 at $\alpha = 0.900$ for push PPR—both chosen with hindsight on the test set, and therefore an upper bound on any constant-α policy. The per-instance rule attains 0.894 and 0.904: margins of +0.217 and +0.168 over the best possible constant. The conventional $\alpha = 0.85$ gives 0.637 and 0.725.

The interpretation is direct. Our instance pool spans effective densities from near-isolation to near-complete connectivity, the optimal α moves across most of its range in response, and no

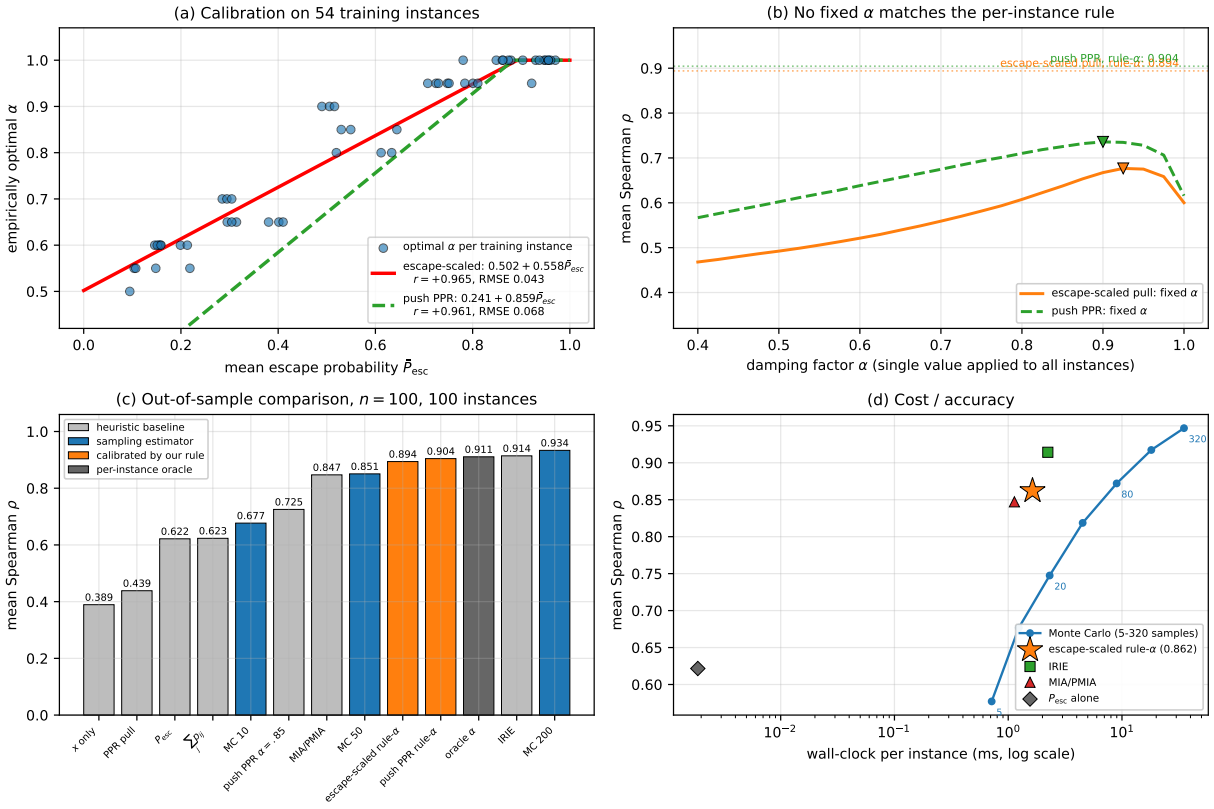


Figure 1: **(a)** The calibration, fitted independently for two operator families. **(b)** Out-of-sample mean ρ as a function of a single fixed α applied to all instances; triangles mark the best attainable constant, dotted lines the per-instance rule. **(c)** Comparison against influence-ranking baselines at $n = 100$. **(d)** Accuracy against wall clock.

single value serves both ends. What makes this actionable is that $\bar{P}_{\text{esc}}$ locates an instance on that axis at no cost.

5.2 The rule describes the problem, not one operator

Table 1: The calibration refit independently per operator family on the same 54 instances. "% oracle" is rule performance as a fraction of a per-instance oracle over the same α grid.

operator family	fitted rule	r	α=0.85	rule-α	% oracle
escape-scaled pull (4)	$\text{clip}(0.502 + 0.558\,\bar{P}_{\text{esc}})$	+0.965	0.637	**0.894**	98.2%
push PPR (3)	$\text{clip}(0.241 + 0.859\,\bar{P}_{\text{esc}})$	+0.961	0.725	**0.904**	98.7%
IRIE, row-normalised [31]	$\text{clip}(0.292 + 0.593\,\bar{P}_{\text{esc}})$	+0.767	0.404	0.439	94.8%

For the two teleportation-style diffusions the correlation exceeds +0.96 and the rule recovers over 98% of oracle, with slopes that differ—as they should, since the operators propagate at different rates—but the same near-linear dependence on $\bar{P}_{\text{esc}}$. On Watts–Strogatz instances, a topology family never used in calibration, the rule attains $\rho = 0.998$, 0.994 and 0.977 in the three sparsest regimes. The third row marks the limit of its reach: IRIE's row-normalised variant has a different fixed-point structure and the relationship is materially weaker. We report it rather than omit it.

5.3 Scale, robustness, and where it stops working

Applied unchanged to the 400-instance panel at $n = 900$, the correlation weakens ($r = +0.603$ and +0.719) while the usefulness does not: the rule still recovers 95.5% and 94.7% of oracle and beats the best constant α by +0.067 and +0.083. The reason is measurable rather than rhetorical—the fraction of the α grid within 0.005 of an instance's own optimum grows from 0.147 at $n = 64$ to 0.290 at $n = 900$, so the optimum is 1.97× flatter and its location correspondingly noisier while the penalty for missing it shrinks. Refitting on the $n = 900$ data barely helps (RMSE 0.150 → 0.147).

Sensitivity to the coefficients is asymmetric. Rounding to $\alpha = \text{clip}(0.5 + 0.6\,\bar{P}_{\text{esc}})$ costs 0.009; raising the intercept by 0.10 costs 0.031 and the slope by 0.20 costs 0.038; lowering them costs 0.199 and 0.283. The ρ-versus-α curve is steep below its optimum and flat above it, so a practitioner uncertain of the coefficients should err upward, and the round numbers are a serviceable substitute for the fitted ones.

On the 35 real citation instances, applied without refitting, the rule gives $\rho = 0.788$ against 0.765 for the best constant α selectable with hindsight, and 0.669 against 0.622 for push PPR—so it still beats every fixed α, at 91% and 89% of a per-instance oracle. But the correlation that motivated it does not survive: optimal α correlates with $\bar{P}_{\text{esc}}$ at $r = +0.225$ and +0.623 there. The rule continues to select good damping factors on real graphs largely because it selects *high* ones, and high ones are right on these networks. The synthetic panels show the rule is not reducible to "always choose a high α"—there the best fixed α is itself high (0.925) and still reaches only 0.677 against the rule's 0.894—but on these real networks the upper clip does much of the work, and we say so. We report the calibration as empirically useful and mechanistically unconfirmed outside the synthetic regime.

Ablation: edge weighting. Logarithmic, linear and uniform weights give 0.8940, 0.8931 and 0.8881 under (5). The weighting scheme is not load-bearing.

Per-node damping does not follow from P_{esc}. Optimising the α vector directly by coordinate ascent, and scoring it against an *independent* ground-truth estimate to exclude overfitting, yields +0.062 over the scalar oracle, only 11% of the apparent gain being overfit. But it does not come from P_{esc}: setting $\alpha_i = \text{clip}(c_0 + c_1 S_i)$ with S the neighbourhood-smoothed escape probability at

decay β gives 0.867 at $\beta = 0$, 0.885 at $\beta = 0.5$ and 0.894 at $\beta = 0.9$—monotonically worse the more local it becomes. The diagnostic is the grid family, where all $P_{\text{esc}}(i)$ are nearly equal so a local rule has no variation to exploit, yet the per-node oracle still gains +0.341. The exploitable per-vertex signal is structural, not escape-based.

6 The escape probability against message passing

Since P_{esc} is the zeroth iterate of (2), the honest deterministic competitor is that recursion run properly, not a sampling estimator. We implement it and report the comparison. Products are evaluated in log space and the excluded-neighbour term removed by subtraction, which is stable as $p\nu \to 1$.

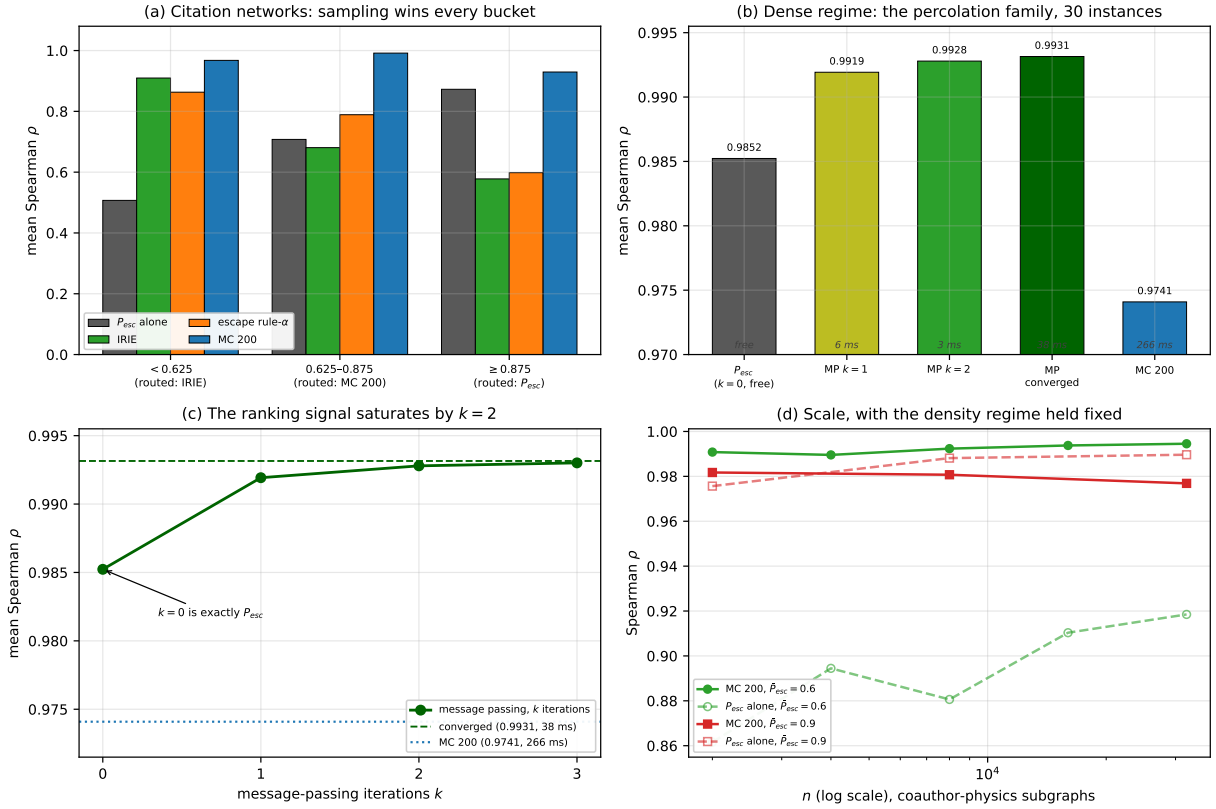


Figure 2: **(a)** On citation networks, sampling wins every bucket of the routing rule (Section 8). **(b)** The deterministic percolation family against sampling on 30 real dense instances; costs beneath each bar. **(c)** The ranking signal as a function of message-passing iterations; $k = 0$ is exactly P_{esc}. **(d)** Size sweep with the density regime held fixed by tuning the edge probability.

On 30 instances drawn from co-purchase and co-authorship networks at $\bar{P}_{\text{esc}} = 0.90$, with n from 10^3 to 1.6×10^4:

method	mean ρ	mean ms	vs P_{esc}	p
P_{esc}, i.e. $k = 0$	0.9852	≈ 0	—	—
message passing, $k = 1$	0.9919	6.0	+0.0067	1.9×10^{-9}
message passing, $k = 2$	0.9928	3.2	+0.0076	1.9×10^{-9}
message passing, converged	**0.9931**	38.0	+0.0079	1.9×10^{-9}
Monte Carlo, 200 samples	0.9741	266.2	−0.0111	4.2×10^{-4}

Three things follow, and only the third is ours.

First, **message passing beats its own $\eta = 1$ limit**, by $+0.0079$ with $p = 1.9 \times 10^{-9}$. Any claim that P_{esc} is the right deterministic score for this regime is therefore wrong, and we do not make it. If three milliseconds are available, iterate.

Second, **the deterministic family beats sampling**, decisively: converged message passing outranks a 200-sample Monte Carlo by $+0.0191$, winning $29/30$ ($p = 9.3 \times 10^{-9}$), at a seventh of the cost. This is the expected consequence of a point Newman makes directly [46]—simulation yields an approximate answer subject to sampling error where message passing yields the value—and we record it as confirmation rather than as a finding.

Third, and this we do offer: **the ranking signal saturates almost immediately**. Two iterations reach within 0.0003 of convergence at a twelfth of the cost, and the free zeroth iterate is already within 0.008. For *ranking* in the dense regime the correlations that message passing corrects are largely rank-preserving. A practitioner needing a score during operator initialisation can use P_{esc} and lose under one percent of a Spearman correlation; one who can afford two sparse sweeps should take them.

We did not test message passing outside the dense regime, where its fixed point is less well behaved on finite graphs, nor against the non-backtracking and collective-influence rankings that occupy the neighbouring literature [55, 45]. We also note that recent work questions what these message-passing solutions identify on graphs with cycles [28].

7 A modern sampling baseline: reverse-reachable sets

The sampling side of the comparison so far is plain possible-world Monte Carlo. The modern sampling machinery for influence, however, is the reverse-reachable-set (RR-set) framework [10, 59, 60], whose current generation—SUBSIM and its successors—improves RR-set construction by up to an order of magnitude and remains the state of the art with guarantees [24, 25, 42]. A comparative study of this problem should include it, so we do.

The estimator. Sample a root $r \sim x$; sample a possible world lazily; the reverse-reachable set of r on an undirected support is $\mathrm{RR}(r) = C(r)$. Then $\mathbb{E}[\,\mathbf{1}\{i \in \mathrm{RR}\}\,] = \sum_r x_r\, P(i \sim r) = y^*_i$ exactly, so the fraction of θ sets containing i is an unbiased estimate of the objective—the singleton-spread special case of RR-set sampling, with the root drawn from the prior as in weighted variants [39]. Our implementation generates sets lazily and level-synchronously: only edges incident to the growing set are examined, each undirected edge at most once, with the Bernoulli draws vectorised over the frontier—the subset-sampling idea behind modern RR-set generators [24, 25] in array form. Correctness was verified against ground truth ($\theta = 6 \times 10^4$ reproduces y^* to within Monte-Carlo noise).

Table 2: The RR-set estimator against possible-world Monte Carlo on all three panels. Same instances and ground truth as the preceding sections; costs are mean wall clock. The rule-α row is the calibrated diffusion of Section 5, for context.

	synthetic, n=100 (100)		citation (35)		dense real (30)	
method	ρ	ms	ρ	ms	ρ	ms
escape-scaled rule-α	0.894	2.2	0.788	13	—	—
Monte Carlo, 200 worlds	0.934	28	0.964	87	0.974	265
RR sets, $\theta = 200$	0.801	31	0.641	99	0.972	556
RR sets, $\theta = 800$	0.924	123	0.738	386	0.989	2251
RR sets, $\theta = 3200$	0.970	493	—	—	—	—

At matched budget, the possible-world estimator wins for whole-graph ranking. On the synthetic hold-out, RR at $\theta = 200$ trails MC at 200 worlds by -0.133 at the same wall clock

($p = 3.9 \times 10^{-16}$, losing 96/100); roughly 16× the sets are needed to draw level. On the citation networks the gap is -0.323 ($p = 2.9 \times 10^{-10}$), driven by a collapse in the sparse bucket (RR-200 mean $\rho = 0.409$ against MC-200's 0.968 over 20 instances). The cause is information per unit of work, and it is structural rather than an implementation artefact. One sampled world yields component labels for *every* vertex simultaneously, so all n estimates share it; one RR set yields a single containment bit per vertex, and in sparse regimes—where components are small—most vertices appear in no set at all after hundreds of draws and are left tied at zero. RR sets are the right primitive for what they were designed for, top-k seed selection with (ε, δ) guarantees on $\sigma(S)$; for ranking *all* vertices they are dominated at matched budget.

The dense regime inverts the picture, and the deterministic family wins it anyway. At $\bar{P}_{\text{esc}} = 0.90$ each RR set is nearly the whole giant component, so per-set information is high: RR-200 ties MC-200 (0.972 vs 0.974) and RR-800 beats it (+0.015, winning 30/30, $p = 1.9 \times 10^{-9}$) at 8.5× its cost. But RR-800 does not significantly beat the free zeroth iterate P_{esc} (+0.004, $p = 0.34$) at four orders of magnitude more cost, and converged message passing (Section 6) outranks it outright at 1/59 of its wall clock (0.9931 at 38 ms against 0.9889 at 2251 ms). In the one regime where RR sets are informative per unit cost, deterministic percolation is better still.

Positioning among modern estimators. Learning-based influence estimators are the other modern family [42]: DeepIS propagates Independent-Cascade probabilities in a PPNP-like architecture [64], MONSTOR and GLIE learn inductive GNN surrogates of the spread [37, 49, 43, 19], and deep reinforcement learning attacks seed selection directly [40, 15]. We cite their own reported costs rather than estimate any: GLIE trains on a 12 GB GPU with grid-searched hyperparameters and reports reliable generalisation to graphs up to roughly ten times its training size [49]; DeepIM reports training times of order 10^3 seconds on a 50,000-node graph and out-of-memory failures for the RL-based PIANO and ToupleGDD on its largest datasets [43]; ToupleGDD trains on a 16 GB V100 [15]. None of this is prohibitive, but all of it is infrastructure—labelled training data (typically generated by Monte-Carlo simulation), a GPU, and a tuning protocol—and these methods target seed-*set* maximisation rather than whole-graph single-seed ranking, so we cite them for context rather than run them. One structural connection is worth recording: the one-hop activation expression $1 - \prod(1 - p)$ at the base of GLIE's first message-passing layer [49] is exactly the escape probability of Section 2. The calibration of Section 3 is therefore not a competitor to trained estimators but the zero-parameter end of the same design space: a closed-form, CPU-only, $O(|E|)$ default for the diffusion family they are benchmarked against, in a query-processing practice that, as recent work notes, still relies primarily on Monte-Carlo sampling and heuristics [62], with RR-set construction itself remaining the bottleneck in adaptive settings [26].

8 A negative result: algorithm selection

No method above wins everywhere on synthetic graphs, and the leader changes systematically with density: IRIE below $\bar{P}_{\text{esc}} \approx 0.63$, Monte Carlo between 0.63 and 0.88, and P_{esc} used directly above that. Since $\bar{P}_{\text{esc}}$ is free and available before any algorithm runs, an obvious idea is to route on it. We tried, and report the outcome because it is negative.

On the 400-instance synthetic panel a three-bucket rule—IRIE below 0.625, Monte Carlo to 0.875, P_{esc} above—reaches $\rho = 0.9477$ under 5-fold cross-validation against 0.9166 for the best single method ($p = 0.009$), capturing 59% of the gain a per-instance oracle selector would obtain. Three buckets suffice; eight do no better and sixteen begin to overfit. Thresholds fitted at $n = 900$ transfer to $n = 100$ unchanged (0.9524 against 0.9527 fitted natively).

On real citation networks it fails. Over 35 instances the router scores 0.9176 against 0.9637 for simply running a 200-sample Monte Carlo everywhere. Figure 2(a) shows why: sampling wins *every* bucket—0.968 against IRIE's 0.910 in the sparse bucket, 0.992 in the middle, 0.929

against P_{esc}'s 0.873 in the dense one. Refitting the bucket assignments on real data yields exactly 0.9637: the refit router selects sampling everywhere. There is no regime structure left to exploit.

We draw the conclusion the evidence supports. The premise that no method wins everywhere and that a cheap global density statistic says which does is a property of the four generative families, not a general fact. Per-instance algorithm selection from cheap features is a mature paradigm [56, 38], and this is a small instance of it that does not generalise—plausibly because a single scalar cannot express the degree heterogeneity and clustering that distinguish real topologies from synthetic ones. Conclusions about which method leads where, drawn from generative topologies, should not be extrapolated.

9 Limitations

- **The rule is empirical.** Equation (5) has no derivation. It is a two-parameter fit, validated out of sample, across topology, across operator family and across scale, and should be read as such. That $\bar{P}_{\text{esc}}$ is essentially a density statistic suggests a mixing-time or percolation-threshold account may exist; we have not found one, and the collapse of the correlation on real networks means we cannot claim the mechanism is understood.
- **Not every operator.** The relationship is strong for teleportation-style damped diffusions ($r > 0.96$) and materially weaker for IRIE's row-normalised variant ($r = +0.767$). We do not claim it holds for arbitrary propagation schemes, and we have not tested it on message-passing operators.
- P_{esc} **is not ours and is not the best deterministic score.** Section 6: it is the $\eta = 1$ iterate of [32], and iterating beats it. We use it as a free feature, not as a method.
- **Dense regimes do not need diffusion at all.** Where $\bar{P}_{\text{esc}}$ is high, percolation estimates outrank every diffusion we tested. A practitioner should treat $\bar{P}_{\text{esc}}$ as a model-selection signal as well as a damping parameter—subject to the failure documented in Section 8.
- **Assigned probabilities, three genres.** No dataset here carries observed p_{ij}, so we assign them as the influence literature does. Whether any of these results hold under empirically estimated probabilities is untested, as is generalisation beyond citation, co-purchase and co-authorship graphs.
- **Scale.** Synthetic panels reach $n = 900$ and real ones $n = 3.4 \times 10^4$. Behaviour at 10^6 vertices is untested.
- **Modern baselines are scoped, not exhaustive.** The RR-set estimator is benchmarked in its singleton-spread form with lazy subset-sampled generation; we did not integrate the full SUBSIM/HIST code paths [24, 25], whose optimisations target seed-set selection. Trained neural estimators [64, 49, 43, 15, 40] are cited with their own reported costs and limits rather than run: they solve seed-set maximisation, require training infrastructure, and their published generalisation ceilings (Section 7) make a fair zero-shot comparison on our panels a study of its own.
- **Ranking, not estimation.** We evaluate rank correlation only. None of these methods estimates y^* numerically and their output is not calibrated to its scale.

10 Related work

The objective. Expected reachable mass in an independent-edge uncertain graph is single-seed node-weighted influence spread under the Independent Cascade model [34] and, in the uniform case, Edge Percolated Component centrality [44, 20]. The node-weighted form corresponds to targeted

and topic-aware influence maximisation [39, 4]. Diffusion and centrality on uncertain graphs have an established literature [51, 14], as does uncertain-graph query processing [53, 29]; Banerjee [6] surveys uncertain-graph mining broadly. The area remains active: reliability maximisation by edge addition [33], and index structures for reliability and expected-reliable-distance queries [62], whose authors note that current uncertain-graph query methods still rely primarily on Monte-Carlo sampling and heuristics—the practice our comparisons quantify.

Estimating it. Cheap proxies include degree discount [17], MIA/PMIA [16] and IRIE [31]; reverse-reachable-set sampling [10] and its refinements [59, 60], with sketching [21], give near-linear time estimation with guarantees. The RR-set line remains the state of the art: SUBSIM and HIST improve set generation by up to an order of magnitude [24, 25], incremental RR-set maintenance addresses adaptive settings [26], and scale-oriented variants continue to appear at the same venues [22]. Benchmarking studies report no single dominant heuristic [3, 48], a conclusion our synthetic panel reproduces and our real-network results qualify.

Learning-based estimation. A parallel line learns the spread: DeepIS propagates Independent-Cascade probabilities through a PPNP-style architecture [64], MONSTOR and GLIE train inductive GNN surrogates [37, 49], DeepIM learns a generative surrogate for seed-set optimisation [43, 19], and PIANO and ToupleGDD apply deep reinforcement learning [40, 15]; Li et al. [42] survey the field. These methods trade training infrastructure and a tuning protocol for accuracy at inference; Section 7 positions our zero-parameter calibration relative to them, and notes that the one-hop activation at the base of GLIE's architecture is exactly the escape probability.

Percolation. The probability that a vertex belongs to the giant cluster is classical: the $1 - u^k$ generating-function result of Newman, Strogatz and Watts [47] and the degree-dependent bond percolation of Callaway et al. [12]. Karrer, Newman and Zdeborová [32] give the per-node message-passing form of which P_{esc} is the $\eta = 1$ iterate; Newman [46] surveys the method and states its advantage over simulation. Message passing on loopy networks, the situation of our clustered real graphs, is treated in [13]. Percolation quantities have been used for node ranking, though in different limits: non-backtracking centrality at criticality [55] and the collective-influence heuristic for optimal percolation [45]. Hiraoka [28] questions what these solutions identify on networks with cycles. The one-hop expression $1 - \prod(1 - p)$ also appears as the non-isolation probability in wireless connectivity [58].

Diffusion operators. Equation (3) is personalised PageRank [50, 27]; the pull orientation and the sub-stochastic treatment of leaking mass are discussed in [23], and reverse PageRank in [7]. Walk operators with node-dependent absorption are classical [63], and (4) is an instance in which absorption is the escape probability. Personalised PageRank has been carried to uncertain graphs with mutually exclusive edge sets [35]. Efficient PPR computation is a field of its own, surveyed in [65]; that literature accelerates the solve for a *given* α and is orthogonal to selecting one.

Choosing the damping factor. Boldi, Santini and Vigna [9] analyse PageRank as a function of α without proposing a selection rule; Bressan and Peserico [11] show worst-case instability alongside empirical stability. Andersen, Chung and Lang [2] tie α to a target conductance. The closest analytical precedents to (5) derive an optimal α from the parameters of a planted model: Kloumann, Ugander and Kleinberg [36] obtain $\alpha^* = (p_{\text{in}} - p_{\text{out}})/(p_{\text{in}} + p_{\text{out}})$ for seed expansion in a stochastic block model, Avrachenkov, Kadavankandy and Litvak [5] an optimum as an implicit function of the same model's density ratio, and Li, Chien and Milenkovic [41] optimal generalized-PageRank weights under related assumptions. While theoretically estimating such model parameters has been attempted [1], these estimators require known block sizes and behave poorly on non-SBM graphs. Our (5) is an empirical calibration from an observable statistic,

operating without block assumptions, for ranking rather than clustering, and free at initialisation. Tagarelli and Interdonato [57] tested average path length as a damping estimator and reported it did not improve on the default, which is the nearest published negative result to our programme. On the learning side, generalized-PageRank weights are trained end to end in GNNs [18], diffusion coefficients are learned per graph and per class [8], and per-node restart probabilities are learned with supervision [30]; Pirrò [52] adapts the restart *distribution* from a global heterophily statistic while keeping α fixed. All of these are trained or supervised where ours is closed-form and free.

Algorithm selection. Choosing an algorithm per instance from cheap features was posed by Rice [56] and surveyed by Kotthoff [38]. Section 8 is a negative instance of that paradigm rather than a contribution to it.

11 Conclusion

The damping factor in diffusion-based reachability ranking on uncertain graphs is usually inherited or grid searched. It need be neither: the mean escape probability, available for free when the transition operator is built, predicts the best-performing α at $r > 0.96$ for teleportation-style diffusions, and the resulting rule recovers over 98% of a per-instance oracle while beating every constant α, including the best selectable in hindsight. It refits cleanly to a second operator family with a different slope and the same near-linear form, survives a 14× change of scale, tolerates rounding, and still beats every constant on real networks—though the correlation underlying it does not survive there, which we report as an open problem rather than a detail.

Three secondary results bound the surrounding claims. The escape probability is not a new score but the correlation-free iterate of message-passing percolation, and iterating beats it; what is worth recording is that for ranking in the dense regime the signal saturates after two sweeps, so the free iterate is within one percent of convergence. The modern reverse-reachable-set estimator, benchmarked in its singleton-spread form, trails plain possible-world sampling for whole-graph ranking at matched budget—it is the right primitive for top-k seed selection, not for this objective—and in the dense regime where it becomes competitive, deterministic percolation beats it at a fraction of the cost. And routing among heuristics by the same statistic, which works on generative topologies, fails on real ones—evidence that a single global density feature does not carry enough information for algorithm selection here.

The natural next step is a derivation. $\bar{P}_{\mathrm{esc}}$ is a density statistic and the optimal teleportation rate is a mixing-time quantity; an argument connecting them would turn a useful fit into a result, and would predict whether the correlation should be expected to weaken on heterogeneous real topologies as we observe it to.

Reproducibility

Every experiment—the identity checks, the calibration, the cross-operator transfer, the α sweep, the stratified 400-instance panel, the per-node ceiling with its independent-ground-truth control, the message-passing comparison, the reverse-reachable-set baseline on all three panels, the real-network panels and the router cross-validation—is reproducible from the accompanying notebook. Long panels are checkpointed per instance and resumable. Real network data is fetched from GitHub-hosted mirrors of the Planetoid and gnn-benchmark releases.

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The experimental design, ablations and statistical testing reported here were carried out with the assistance of an AI system. The escape-probability formulation and the underlying model are the author's.

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